

Activity 23 Instruction Manual

Activity 23 | Topic 03 | C013 | Version v1.0

Purpose

Earned value is the one technique that tells you objectively whether you are ahead or behind, over or under. You will compute it by hand first, then test whether GenAI gets it right.

Files used

- activity-guide.md and activity-guide.pdf - brief and source instructions
- prompts.pdf - reusable prompt and verification checklist
- mock-data.xlsx - safe fictional project data for practice

Expected output

A complete earned value analysis with variances, indices and a forecast.

Before you begin

1. Form the required group and assign one person to capture evidence.
2. Open mock-data.xlsx and review the title, note and column headings.
3. Open prompts.pdf beside your chosen GenAI tool. Do not upload confidential workplace data.

Detailed procedure

1. Work this classic scenario by hand, individually. Task: develop and install ten printer drivers. Budget: \$100,000 (\$10K per driver). Time: 10 weeks (1 driver per week). At week 5: 4 drivers are complete and \$47,500 has been spent.
2. Compute the three base values. PV = the budget for work SCHEDULED by now. EV = the budget for work ACTUALLY DONE. AC = what has actually been spent.
3. Confirm your answers: PV = \$50,000 (5 drivers scheduled x \$10K). EV = \$40,000 (4 drivers done x \$10K). AC = \$47,500.
4. Compute the variances. Cost Variance CV = EV - AC = \$40,000 - \$47,500 = -\$7,500 (negative is bad - over budget). Schedule Variance SV = EV - PV = \$40,000 - \$50,000 = -\$10,000 (negative is bad - behind schedule).
5. Compute the indices. CPI = EV / AC = 40,000 / 47,500 = 0.84. SPI = EV / PV = 40,000 / 50,000 = 0.80. Both below 1 - over budget and behind schedule.
6. Forecast the outcome. EAC = BAC / CPI = 100,000 / 0.84 = approximately \$119,000. ETC = EAC - AC. State what you would tell the sponsor.
7. NOW test GenAI. Give it the same scenario and ask for PV, EV, AC, CV, SV, CPI, SPI and EAC. Compare every figure against yours.
8. Where it differs, identify the error. GenAI commonly confuses EV with AC, or computes EAC without using CPI. This is precisely why you must be able to do this by hand.
9. Discuss as a group: the assessment is open book, but the arithmetic is yours. What is the minimum you must be able to compute without help?

Use the supplied mock data

Copy only the rows needed for the current step and tell the model the data is fictional. If the model changes an identifier or number, reject the change and rerun with an explicit preserve-values instruction.

Quality control

- You can compute PV, EV, AC, CV, SV, CPI, SPI and EAC unaided for a fresh scenario, and you found at least one error in the AI's version.
- Compare the first AI draft with the final human-corrected artifact and retain both as evidence.
- Label assumptions, decisions, owners and dates so another project manager can audit the result.

Troubleshooting

If the response is generic, add the project objective, constraints, audience and required columns. If arithmetic is inconsistent, request formulas and substitutions, then calculate independently. If a source cannot be opened, remove the claim or replace it with verified evidence.

Submission evidence

Submit the completed project artifact, the final prompt, a short critique of the AI output, and any screenshot or exported file required by the activity.